

1. Declaration

I, [Student Name], declare that this assignment, titled [Assignment Title], is my own original work and has not been copied from any other source except where explicitly acknowledged. I have not engaged in plagiarism, collusion, or any other form of academic misconduct in the preparation and submission of this assignment. All sources of information and data used in this assignment have been properly cited and referenced in accordance with the prescribed guidelines. I have not used unauthorized assistance in the preparation of this assignment and have not allowed any other student to copy my work. I am aware that any breach of academic integrity may result in disciplinary action as per the [policies of Monash University](#), which may include failing this assignment or the course, and further academic penalties.

Signature: Shashank

Date: 14 September, 2025

2. Github Check

Enter your Github details here.

Github Username <i>Enter your username here</i>	only-bugs
Repository Shared? <i>Have you started and shared your assignment repository with your tutor yet?</i>	https://github.com/Only-Bugs/Nutri-buddy/tree/A1.3/Business_Requirements_C

3. Self-Evaluation

Rate your performance for each criteria. Put a ☒ (tick) in the box where you think your work belongs.

Criteria	Exceeds Expectations	Meets Expectations	Needs Improvement	Fail to meet expectations
BR (C.1): Authentication		☒		
BR (C.2): Role-based authentication	☒			
BR (C.3): Rating	☒			
BR (C.4): Security	☒			

4. Screen Recording of BRs

Create a 3 minute video showing your basic web application in action! Upload this video to your Google Drive and put the link here (ensuring that you have updated the access list so its not private).

 A1.3_business Requirements_C.mp4

5. Reflections: Implementation of C.4 Security

If you have implemented BR C.4, in less than 200 words describe the approach that you have taken to implementing Security in your application. What security flaws were you trying to prevent and what security measures have you implemented to fix those flaws? How do you know that these measures will help prevent those issues from happening? Optionally you can cite external sources to provide evidence for your claim.

Security was implemented with a focus on preventing XSS attacks and ensuring safe handling of sensitive data. On the frontend, all user inputs in the login and registration forms are first validated using **Vee-Validate/Yup** and then sanitized with **DOMPurify**. This combination ensures malicious payloads, such as `<script>` injections, cannot execute and instead are safely neutralized. On the backend, I integrated **Helmet** to enforce secure HTTP headers and **CORS** to control resource access across origins. Together, these measures reduce the risk of reflected XSS, clickjacking, and other injection-style exploits. Sensitive credentials, such as API keys, are never exposed in the client; instead, they are stored in `.env` files and accessed server-side, which aligns with best practices for secret management.

6. Reflections: Challenges

What has been the most challenging part of this assignment for you? How has this stretched you as a programmer?

The most challenging part of this assignment was balancing functionality with security. Getting authentication and role-based access working was straightforward, but ensuring it was secure pushed me to think differently. I had to research libraries like DOMPurify and Helmet, learn about sanitization and validation, and carefully wire them into both frontend and backend without breaking the user flow. This stretched me as a programmer by making me more mindful of real-world vulnerabilities, not just features. It reminded me that building software isn't only about making it work, but making it safe to use.

7. Declaration: Additional Help

Any tools that you used (including Gen AI or existing code reuse) must be declared here.

Note: GenAI is not allowed for coding purposes in any assignment,

However, you may use GenAI for brainstorming and problem solving. You need to declare all such uses here. One row per help used.

Name	Description
<i>Example: ChatGPT for brainstorming ideas</i>	<i>I used ChatGPT to brainstorm how to do X because I was feeling stuck with Y problem.</i>
Chatgpt for Researching libraries	I have used Chatgpt to assess the different ways the same problem can be solved and the most popular third party libraries.